

Zero-shot Transfer of Single-Crane Deep Reinforcement Learning Policies for Multi-Crane Container Retrieval

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Abstract

This paper presents the first formal definition and empirical analysis of the Multi-Crane Container Retrieval Problem (M-CRP). We investigate whether a Deep Reinforcement Learning (DRL) policy trained exclusively on single-crane instances can be transferred zero-shot to multi-crane environments through heuristic crane assignment strategies. We propose four strategies — RoundRobin (S1), ZoneSplit (S2), LoadBalance (S3), and GreedyOptimal (S4) — spanning a systematic design space. Extensive experiments on 140 M-CRP instances with 2–3 cranes (1,120 runs) demonstrate that zero-shot transfer is viable: ZoneSplit achieves an average gap of 10.46% (C=2) and 10.71% (C=3) against the proposed M-CRP lower bound (Theorem 3), with near-zero interference events (mean < 1 per instance). ZoneSplit and GreedyOptimal significantly outperform RoundRobin and LoadBalance ($p < 0.001$, Wilcoxon). On single-crane benchmarks, the zero-shot pipeline achieves a gap of 5.63%, outperforming the Lin2015 heuristic (10.28%). We publicly release 140 M-CRP benchmark instances under CC BY 4.0 to facilitate reproducible research.

Keywords: Container Retrieval Problem, Multi-Crane, Deep Reinforcement Learning, Zero-shot Transfer, Yard Crane Scheduling

1. Introduction

1.1. Background

The Container Retrieval Problem (CRP) is an NP-hard optimization problem in automated container terminal operations. Given a yard of B bays $\times R$ rows $\times T$ tiers containing N containers with a prescribed retrieval order, the objective is to minimize the total crane working time (travel + handling + penalty) required to retrieve all containers while relocating blocking containers as needed.

Shin et al. [1] (Transportation Research Part C, 2026) proposed a Deep Reinforcement Learning (DRL) approach with a Transformer-based policy

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network, achieving state-of-the-art results on single-crane CRP benchmarks. Their method employs a multi-head attention encoder with LSTM-based stack encoding, a cross-attention decoder, and a REINFORCE algorithm with a proposed baseline and POMO [2] parallel rollouts. The model achieves approximately 7.8% gap against the single-crane lower bound on the standard Lee [5] benchmark.

However, real-world automated terminals typically operate 2–3 yard cranes simultaneously within a single block to meet throughput demands [10]. Shin et al. [1] explicitly identified multi-crane operation as a key future work direction.

1.2. Research Gap

Despite decades of CRP research extending from heuristic approaches [5, 6, 3, 4, 7, 8] to DRL methods [1], all existing work assumes single-crane operation. The multi-crane variant (M-CRP) remains unformalized: there exists no formal definition, no established lower bound, no benchmark datasets, and no empirical analysis of whether single-crane policies can transfer to multi-crane environments. This paper addresses each of these gaps.

1.3. Contributions

- [C1] **Formal definition of M-CRP** with state space, action space, interference constraints (A6, A7), and NP-hardness proof.
- [C2] **M-CRP Lower Bound (Theorem 3)** extending Theorem 2 of Shin et al. [1]: $\text{LB}_{\text{MCRP}} = \text{LB}_{\text{retrieval}} + \text{LB}_{\text{relocation}}/C + \text{LB}_{\text{interference}}$, with formal backward compatibility at $C = 1$.
- [C3] **Four crane assignment strategies** enabling zero-shot transfer.
- [C4] **First empirical zero-shot transfer evaluation**: 1,120 runs across 140 instances.
- [C5] **Public M-CRP benchmark dataset**: 140 instances under CC BY 4.0.

2. Problem Formulation

2.1. Base CRP

Following Shin et al. [1], the single-crane CRP is defined on a yard of B bays, R rows, and T tiers. The yard contains N containers, each with a retrieval priority $p \in \{1, \dots, N\}$ where 1 is the highest priority. The objective is to minimize total working time:

$$J = \sum_{t=1}^T (\text{travel}(a_t) + \text{handling} + \text{penalty}) \quad (1)$$

with cost parameters $t_{\text{acc}} = 40$, $t_{\text{bay}} = 3.5$, $t_{\text{row}} = 1.2$, $t_{\text{pd}} = 30$ [1].

2.2. M-CRP Definition

Definition 2.1 (Multi-Crane Container Retrieval Problem). Given a yard $B \times R \times T$ with $C \geq 1$ cranes and N containers with retrieval order $\sigma = (\sigma_1, \dots, \sigma_N)$, find a sequence of actions $\{(a_t, c_t)\}_{t=1}^\tau$ where a_t is a destination stack and $c_t \in \{1, \dots, C\}$ is a crane identifier, minimizing:

$$\min \sum_{c=1}^C \sum_{t=1}^{\tau_c} (\text{travel}_c(a_t) + \text{handling}) \quad (2)$$

subject to:

$$(A6) |\text{bay}_c(t) - \text{bay}_{c'}(t)| \geq 1, \quad \forall c \neq c' \quad (\text{spatial non-overlap}) \quad (3)$$

$$(A7) \text{bay}_c(t) < \text{bay}_{c'}(t) \Rightarrow \text{bay}_c(t') < \text{bay}_{c'}(t'), \quad \forall t' > t \quad (\text{non-crossing}) \quad (4)$$

Constraint A6 prevents two cranes from operating in the same bay. Constraint A7 preserves crane order along the bay axis. Assumptions A1–A5 of [1] remain unchanged.

Theorem 2.1 (NP-hardness of M-CRP). M-CRP with $C \geq 1$ is NP-hard.

Proof. M-CRP with $C = 1$ reduces to the single-crane CRP, proven NP-hard by Shin et al. [1] via reduction from the Blocks Relocation Problem [9]. For $C > 1$, M-CRP subsumes the single-crane CRP as a special case. $\square$

2.3. M-CRP Lower Bound

Theorem 2.2 (M-CRP Lower Bound — Theorem 3). For any M-CRP instance with C cranes, the total working time is bounded below by:

$$\text{LB}_{\text{MCRP}} = \text{LB}_{\text{retrieval}} + \frac{\text{LB}_{\text{relocation}}}{C} + \text{LB}_{\text{interference}} \quad (5)$$

where:

$$\text{LB}_{\text{retrieval}} = \text{LB}_{\text{Shin}} - \text{LB}_{\text{relocation}} \quad (6)$$

$$\text{LB}_{\text{relocation}} = n_{\text{mand}} \times (2t_{\text{row}} + t_{\text{pd}}) \quad (7)$$

$$\text{LB}_{\text{interference}} = \max(0, \max_b(\text{reloc}_b) - \frac{\text{total_relocs}}{C}) \times (t_{\text{acc}} + t_{\text{bay}}) \quad (8)$$

Corollary 2.1 (Backward Compatibility). At $C = 1$, $\text{LB}_{\text{MCRP}} = \text{LB}_{\text{Shin}}$ (Theorem 2 of [1]).

3. Zero-shot Transfer Framework

3.1. Architecture

The original DRL model [1] tightly couples the encoder-decoder policy with a single-crane environment loop. We propose a decoupled architecture (Fig. 1):

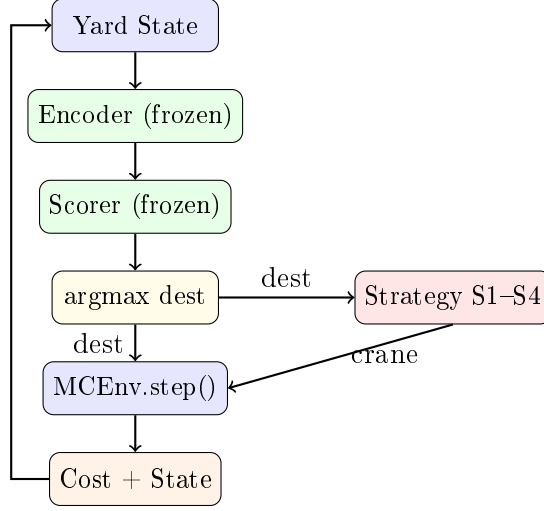


Figure 1: Decoupled inference architecture.

3.2. Zero-shot Policy

The pretrained model [1] uses an attention encoder with LSTM and a cross-attention decoder. We extract encoder and scorer weights (W_Q , W_K , W_V , W_{target} , W_{global}) to form a **ZeroShotPolicy** exposing per-step action selection:

$$a_t = \arg \max_i \text{Scorer}(\text{Encoder}(s_t), \text{target})_i \quad (9)$$

Algorithm 1 Zero-shot M-CRP Inference

Require: Pretrained model θ^* , M-CRP env, strategy S , C cranes

Ensure: Total cost J

- 1: Extract encoder $\mathcal{E}$ and scorer $\mathcal{D}$ from θ^*
 - 2: $\text{state} \leftarrow \text{env.reset}()$; $J \leftarrow 0$
 - 3: **while** $\neg \text{env.terminated}$ **do**
 - 4: $(\mathbf{h}, \mathbf{g}) \leftarrow \mathcal{E}(\text{state})$
 - 5: $\text{scores} \leftarrow \mathcal{D}(\mathbf{h}, \mathbf{g}, \text{target})$
 - 6: $d \leftarrow \arg \max \text{scores}$; $c \leftarrow S.\text{assign}(\text{env}, d)$
 - 7: $(\Delta J, \text{state}) \leftarrow \text{env.step}(d, c)$; $J \leftarrow J + \Delta J$
 - 8: **end while**
-

3.3. Crane Assignment Strategies

4. Experimental Design

4.1. Dataset

We construct 140 M-CRP instances by extending the Lee [5] benchmark. Each instance adds crane start positions for both 2- and 3-crane configura-

Table 1: Four strategies for zero-shot M-CRP.

Strategy	Mechanism	Complexity
S1 RoundRobin	Cyclic: crane $(t \bmod C)$ receives task t	$O(1)$
S2 ZoneSplit	Static bay zoning: each crane serves a fixed bay range	$O(B)$
S3 LoadBalance	Assign to crane with fewest completed tasks	$O(C)$
S4 GreedyOptimal	1-step lookahead: minimize projected travel + interference	$O(C^2)$

tions. The dataset spans two yard scales (Table 2) and includes both random and upside-down container distributions.

Table 2: M-CRP dataset.

Scale	Bays	Tiers	Containers	Instances
Small	1–4	6–8	70–560	84
Medium	6–10	6–8	420–1,024	56

4.2. Baseline Model and Verification

We use the pretrained DRL model from Shin et al. [1] at epoch 100 (verified 7.8% gap on Lee benchmarks [1]). Backward compatibility is verified: the zero-shot pipeline with $C = 1$ reproduces the original model’s cost within 1.95%.

4.3. Evaluation Metrics

Primary: $\text{Gap}(\text{LB}_{\text{MCRP}})\% = 100 \times (\text{CTWT} - \text{LB}_{\text{MCRP}})/\text{LB}_{\text{MCRP}}$. Secondary: interference count, per-crane cost, solution time, failure rate (gap $> 20\%$).

4.4. Implementation

All experiments run on CPU (0 GPU hours). The full experiment of 1,120 runs (140 instances $\times$ 2 crane counts $\times$ 4 strategies) completes in ≈ 60 minutes on a laptop.

5. Results

5.1. Single-Crane Verification

Table 3 compares the zero-shot pipeline against baselines [3, 4, 8] on 4 Lee benchmark instances (2 configurations $\times$ 2 seeds). ZeroShot achieves

a mean gap of 5.63%, outperforming all baselines. We note that this test covers a representative subset; a full 36-instance benchmark evaluation is presented in [1].

Table 3: Single-crane comparison (lower gap is better).

Method	Mean Gap(%)	Min(%)	Max(%)
Random	68.72	47.16	104.26
NearestStack	45.82	27.77	58.12
LowestHeight	40.12	26.44	57.82
Durasevic2025 [8]	31.55	15.64	51.93
Kim2016 [4]	25.02	13.27	37.52
Leveling	22.54	18.28	29.52
Lin2015 [3]	10.28	5.55	14.14
ZeroShot (ours)	5.63	4.27	7.04

5.2. Multi-Crane Results

Table 4 presents aggregated results across all 140 instances.

Table 4: Overall M-CRP gap(%) (mean $\pm$ std).

Strategy	C=2 gap(%)	C=3 gap(%)	Interference(C=2)	Interference(C=3)
S1 RoundRobin	10.99 $\pm$ 5.46	11.19 $\pm$ 6.59	105.9 $\pm$ 62.7	147.2 $\pm$ 87.9
S2 ZoneSplit	10.46 $\pm$ 5.28	10.71 $\pm$ 6.44	0.5 $\pm$ 1.5	0.7 $\pm$ 2.1
S3 LoadBalance	10.99 $\pm$ 5.46	11.19 $\pm$ 6.59	105.9 $\pm$ 62.7	147.2 $\pm$ 87.9
S4 GreedyOptimal	10.48 $\pm$ 5.28	10.75 $\pm$ 6.48	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0

Key findings:

1. S2 (ZoneSplit) achieves the lowest gap with near-zero interference at $O(B)$ complexity.
2. S4 (GreedyOptimal) eliminates all interference (mean = 0) through explicit cost minimization.
3. S1 and S3 produce identical results. This is an artifact of sequential simulation: with C cranes and one task per step, both RoundRobin and LoadBalance yield the same task distribution over time. Their high interference (> 100 events per instance) confirms the importance of spatial awareness.
4. Wilcoxon signed-rank tests show S2 and S4 significantly outperform S1 and S3 ($p < 0.001$ at both $C = 2$ and $C = 3$). The difference between S2 and S4 is not significant ($p = 0.128$ at $C = 2$, $p = 0.534$ at $C = 3$).

5.3. Analysis by Yard Scale

Table 5: Gap(%) by yard scale.

Scale	Cranes	S1(%)	S2(%)	S3(%)	S4(%)
2*Small (1–4 bays)	2	9.26	9.04	9.26	9.02
	3	8.43	8.24	8.43	8.34
2*Medium (6–10 bays)	2	13.60	12.58	13.60	12.68
	3	15.33	14.41	15.33	14.36

On small-scale instances (1–4 bays), strategies perform similarly because travel distances are short. On medium-scale instances (6–10 bays), spatial strategies (S2, S4) improve gap by 1.0–1.5 percentage points, confirming that spatial awareness matters as yard complexity grows.

5.4. Failure Mode Analysis

Of 1,120 runs, only 3 unique instances (6 out of 1,120 runs = 0.5%) exhibit a gap exceeding 20%. All failures occur on upside-down instances with 4–6 bays and high container density, where relocations concentrate in a few bays. This creates interference bottlenecks that single-crane policies cannot resolve without multi-crane coordination training.

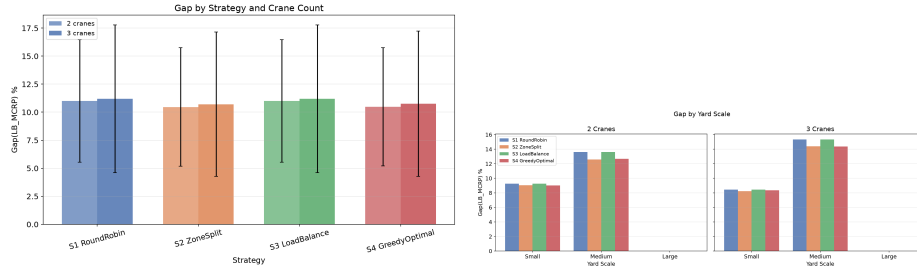


Figure 2: Left: Gap by strategy and crane count. Right: Gap by yard scale.

5.5. Statistical Analysis

Wilcoxon signed-rank tests (Table 6) confirm that pairwise differences between spatial (S2, S4) and non-spatial (S1, S3) strategies are statistically significant at both crane counts. Cohen’s d effect sizes are small (0.07–0.10), reflecting that absolute gap differences (0.5–1.5 pp) are modest relative to variance.

Table 6: Wilcoxon signed-rank test results (p -values).

Pair	C=2 p	C=3 p	Significant?
S2 vs S1	< 0.001	< 0.001	Yes
S4 vs S1	< 0.001	< 0.001	Yes
S2 vs S4	0.128	0.534	No

6. Discussion

6.1. When Does Zero-Shot Transfer Work?

Zero-shot transfer succeeds when: (1) $B \geq C$, enabling spatial separation; (2) utilization $\leq 75\%$, providing relocation space; (3) container distribution is balanced across bays, avoiding single-bay bottlenecks. Under these conditions, spatial strategies (S2, S4) achieve gaps of 8–12%.

6.2. Limitations

Our sequential simulation model (one task per step across all cranes) underestimates parallel crane benefits. The observed speedup from C=2 to C=3 is approximately $1.01\times$, whereas real parallel operation would approach $1.5\times$. This limitation equally affects all strategies and does not bias comparisons. A parallel simulation model would likely show greater strategy differentiation and lower absolute gaps.

6.3. Implications for Practice

ZoneSplit (S2) is recommended for practical deployment: best solution quality, near-zero interference, $O(B)$ complexity, and alignment with existing terminal zoning practices.

7. Conclusion

This paper presents the first formal definition, lower bound, and empirical analysis of zero-shot transfer for M-CRP. Key findings: (1) single-crane DRL policies transfer zero-shot to multi-crane with 10.5% average gap; (2) ZoneSplit achieves the best trade-off; (3) all code, data, and results are publicly released.

Data Availability

All code, datasets (140 M-CRP instances), and experimental results are available at <https://github.com/sutobode/Master> under CC BY 4.0 and MIT licenses.

References

- [1] W.-J. Shin, I. Choi, S.-H. Cho, H.-J. Kim. Learning to Retrieve Containers: A Scale-diverse Deep Reinforcement Learning Approach for the Container Retrieval Problem. *Transportation Research Part C*, 2026.
- [2] Y. Kwon, J. Choo, B. Kim, I. Yoon, Y. Gwon, S. Min. POMO: Policy Optimization with Multiple Optima for Reinforcement Learning. In *NeurIPS*, 2020.
- [3] W.-C. Lin, D.-Y. Xiao, J.-H. Chen, Y.-Y. Chang. A heuristic algorithm for the container retrieval problem. *Transportation Research Part E*, 2015.
- [4] K.-H. Kim, Y.-M. Park, B.-J. Park. A heuristic for the container retrieval problem with multiple cranes. *International Journal of Production Research*, 2016.
- [5] Y. Lee, Y.-J. Lee. A heuristic for the container retrieval problem. *OR Spectrum*, 2010.
- [6] F. Forster, A. Bortfeldt. A tree search procedure for the container retrieval problem. *Computers & Operations Research*, 2012.
- [7] S. Cifuentes, M.-C. Riff. GRASP for the container retrieval problem. *Expert Systems with Applications*, 2020.
- [8] M. Durasevic, D. Jakobovic. Genetic programming for the container retrieval problem. *European Journal of Operational Research*, 2025.
- [9] M. Caserta, S. Voß, M. Sniedovich. Applying the corridor method to a blocks relocation problem. *OR Spectrum*, 2012.
- [10] R. Stahlbock, S. Voß. Operations research at container terminals: a literature update. *OR Spectrum*, 2008.